LLM-Based Sequential Clinical Diagnosis

Recent Progress and Open Challenges

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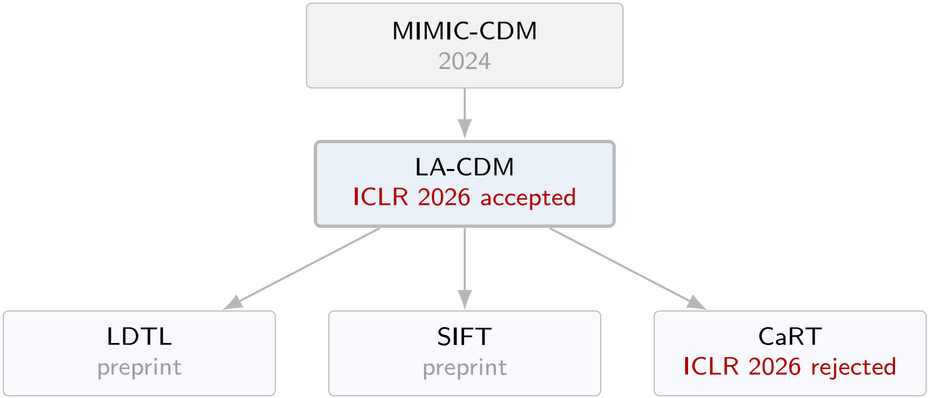
LA-CDM · LDTL · SIFT · CaRT · MIMIC-CDM

Survey Map



Recently Emerging Problem Setting

MIMIC-CDM (2024) introduced **sequential evidence acquisition** as a benchmark, shifting LLM diagnosis from "full-information classification" to learning "which test to inspect and when."

Main Focus

What is a good trajectory?

We have the ground-truth diagnosis label, but **we do not have labels for good diagnostic paths**. Recent studies therefore fill this gap with different proxies, without directly observing the optimal trajectory.

Initial → a → b → c → Diagnosis

MIMIC-CDM Dataset

Benchmark Goal

Instead of a classification task where all patient information is given at once, MIMIC-CDM defines a clinical decision-making benchmark where the model must **select tests sequentially** and then predict the diagnosis.

Scale and Labels

A total of **2,400** abdominal disease patient cases.

Disease list

- appendicitis
- cholecystitis
- diverticulitis
- pancreatitis

Action space: 12 actions

1. Lab tests 7 actions

Complete Blood Count, Basic Metabolic Panel, Comprehensive Metabolic Panel, Renal Function Panel, Liver Function Panel, Urinalysis, Electrolyte Panel

2. Imaging reports 4 actions

CT, MRI, Radiograph, Ultrasound

3. Physical examination 1 action

Some patients do not have records for certain test results.

Language Agents for Hypothesis-driven Clinical Decision Making with Reinforcement Learning

LA-CDM

ICLR 2026

LA-CDM: Motivation

1. Existing clinical RL mostly focuses on tabular data

Prior reinforcement-learning approaches to clinical decision-making are mostly designed for **tabular data**. As a result, they miss clinically important modalities such as clinical notes and imaging reports.

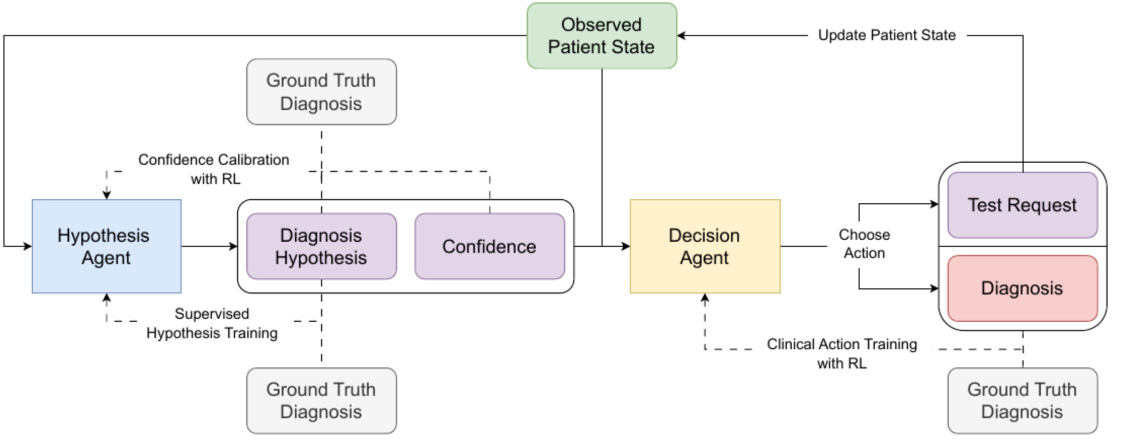
2. Existing LLM clinical decision-making is mostly zero-shot

Existing LLM-based clinical decision-making mostly relies on prompting or zero-shot reasoning. It does not explicitly train **diagnostic test selection or sequential decision-making itself**.

LA-CDM Problem Setting

LA-CDM frames LLM-based clinical decision-making as sequentially requesting tests from a multimodal patient state and making a diagnosis when confidence is sufficient.

LA-CDM: Main Figure

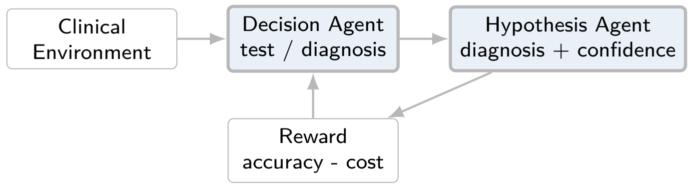


Components

Hypothesis: maintains possible diagnosis candidates based on the patient state observed so far.

Decision agent: requests the next test, or selects the final diagnosis when confidence is sufficient.

LA-CDM: Method



Cyclic training objectives

1. **Hypothesis generation**
   SFT trains HA to generate the ground-truth diagnosis y_{true} at each state.
2. **Uncertainty-awareness**
   GRPO calibration aligns confidence with $J(h_j) = \mathbb{1}[h_j = y_{\text{true}}]$.
3. **Efficient action selection**
   GRPO trains DA to request useful tests and decide when to stop/diagnose.

DA phase reward

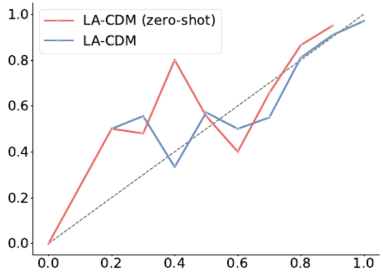
$$R_{\text{diag}} = \begin{cases} r_{\text{pos}} & y_{\text{pred}} = y_{\text{true}} \\ r_{\text{neg}} & y_{\text{pred}} \neq y_{\text{true}} \\ r_{\text{invalid}} & \text{format error} \end{cases}$$

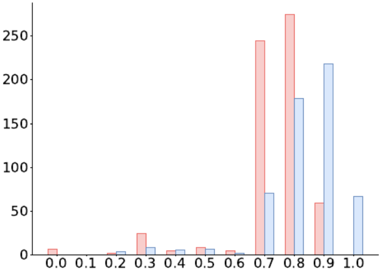
$$R_{\text{cost}}(T) = -\sum_{t_j \in T} c(t_j)$$

This reward is used only in the **efficient action selection phase**. Hypothesis SFT and confidence calibration use separate objectives and are alternated within the training cycle.

LA-CDM: Results and Implications

Method	Accuracy					F1-Score		Avg.
	Append.	Cholec.	Divert.	Pancr.	Mean	Micro	Macro	Test Cost
OASST*	82.0	48.0	45.5	44.1	54.9	-	-	-
SFT-all	98.4	89.8	95.8	87.5	92.8	93.6	92.9	$3792.79
SM-DDPO†	74.3	0.0	15.6	58.0	37.0	45.4	31.8	-
ReAct	90.2	79.7	66.7	62.9	74.9	79.1	74.8	$1480.32
LA-CDM (ZS)	73.5	55.1	72.0	57.5	64.5	65.3	64.5	$1521.73
LA-CDM	**93.1**	**83.6**	**75.0**	**73.5**	**81.3**	**84.1**	**81.3**	**$1295.61**





What the Results Suggest

- LA-CDM mean accuracy 81.3, macro F1 81.3
- Higher performance and lower average test cost than ReAct
- Confidence calibration also improves after training

Uncertainty-Guided Latent Diagnostic Trajectory Learning for Sequential Clinical Diagnosis

LDTL

arXiv 2026

LDTL: Motivation

Terminal reward is insufficient for trajectory learning

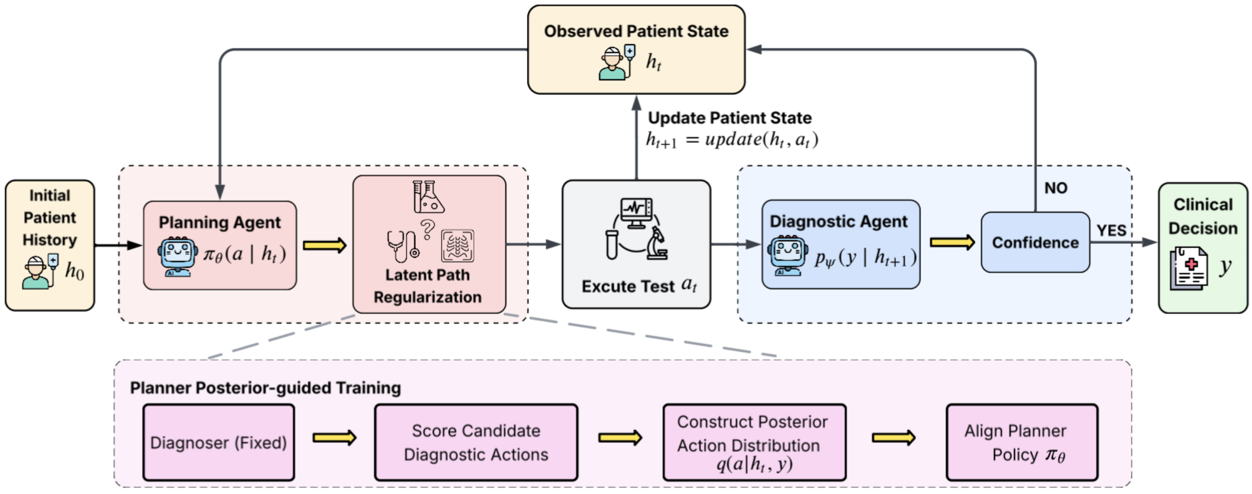
Optimizing performance and cost only through terminal reward does not directly control the **structure or distribution of diagnostic trajectories**. This can be unstable with limited clinical trajectories and provides weak supervision for desirable paths.

Initial → a → b → c → Reward

LDTL Problem Setting

LDTL formulates test selection as a **Latent Diagnostic Trajectory Learning** problem and constructs action signals based on diagnostic improvement rather than simple rewards.

LDTL: Main Figure



Agent Roles

Planning LLM agent: selects the next diagnostic action based on the current patient state.
Diagnostic LLM agent: predicts the disease label and confidence after new evidence is added.

LDTL: ideal vs. approximation

Ideal: trajectory posterior

$$q(z \mid h_0, y) = \frac{\exp(\beta S(z))}{\sum_{z' \in Z(h_0)} \exp(\beta S(z'))}$$

$$q(a_t \mid h_t, y) = \sum_{z : z_t = a_t} q(z \mid h_0, y)$$

trajectory z	$S(z)$	$q(z)$
(A, B, C)	3.0	0.30
(A, C, B)	2.0	0.12
(B, A, C)	2.5	0.20
(B, C, A)	3.5	0.35
(C, A, B)	1.0	0.02
(C, B, A)	0.5	0.01

$$q(A) = 0.42, \quad q(B) = 0.55, \quad q(C) = 0.03$$

A posterior is formed from the full trajectory score $S(z)$, then probabilities of trajectories starting with the same action are aggregated to obtain the action posterior.

Approx.: one-step IG

$$IG(h_0, a) = \log p(y \mid h_0 + a) - \log p(y \mid h_0)$$

$$q(a \mid h_0, y) \propto \exp(IG(h_0, a)/\tau)$$

action	one-step IG
A	0.70
B	0.50
C	0.10

$$IG(A) > IG(B) > IG(C)$$

In practice, training does not enumerate full trajectories. Instead, it forms the action posterior from the diagnostic improvement obtained by adding one action to the current state.

LDTL: Method

Stage 1. Train the diagnostic LLM first

The diagnostic LLM is SFT-trained to predict the correct disease label and output a confidence score when full patient information, i.e., complete diagnostic results, is provided.

Stage 2. Freeze the diagnostic LLM and train the planner

After freezing the trained diagnostic LLM, the planning LLM learns which diagnostic action to select. At each step, candidate actions are evaluated, an action-level posterior $q(a_t \mid h_t, y)$ is constructed from **diagnostic improvement**, and the planner policy is aligned to it.

$$\mathcal{L}_{\text{planner}} = \sum_{t=1}^{T} \mathrm{KL}(q(a \mid h_t, y) \,\|\, \pi_\theta(a \mid h_t))$$

Key Point

Instead of constructing ground-truth trajectories directly, LDTL uses **probability changes from the frozen diagnostic LLM** to create planner supervision.

LDTL: Results and Implications

Method	Acc	Macro-F1	App.	Chol.	Div.	Pan.
RS	84.8	83.7	82.8	85.4	90.4	85.2
w/o LP	88.6	88.5	94.8	90.7	80.7	78.7
LDTL	**93.4**	**91.7**	98.9	95.4	78.8	87.9

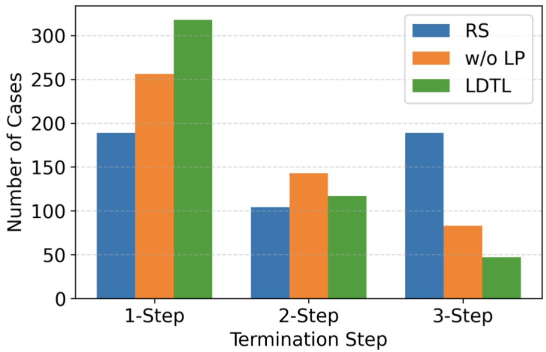


Figure 4: Distribution of diagnostic termination steps

Interpretation

Table 2: LDTL outperforms RS and w/o LP in Acc and Macro-F1. This suggests that latent trajectory supervision improves final diagnostic performance.

Figure 4: LDTL has more 1-step terminations and fewer 3-step terminations. This suggests that it learns to gather necessary information earlier and reduce unnecessary extra tests.

Learning What Matters: Criticality-Guided Reinforcement Learning for Sequential Clinical Diagnosis

SIFT

Preprint 2026

SIFT: Motivation

1. Existing benchmarks restrict actions and diagnoses to closed spaces

AMIE, ED-Copilot, and LA-CDM-style benchmarks restrict the action space to a predefined test menu and the diagnosis space to a fixed label set. Real clinical reasoning, however, is closer to open-ended interaction.

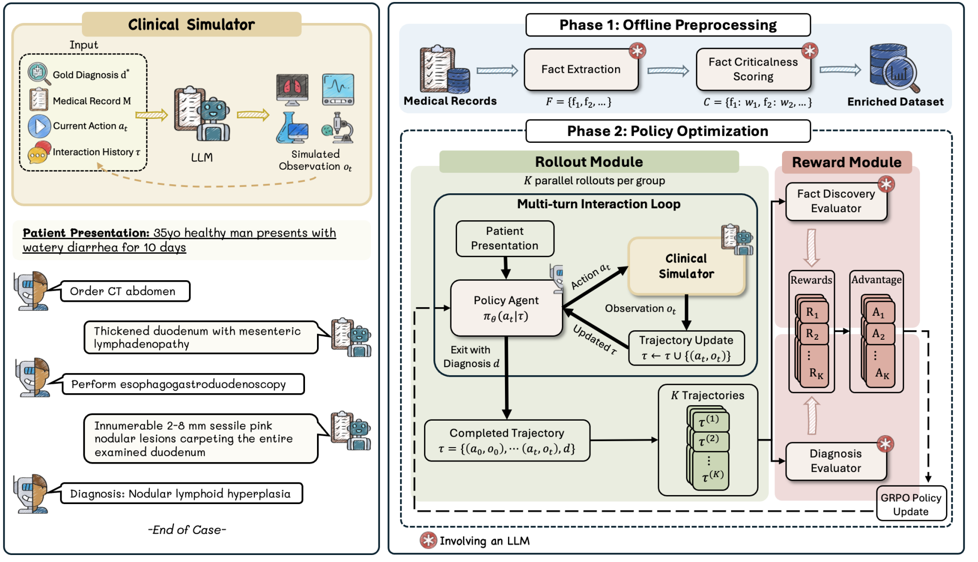
2. Terminal reward does not capture evidence quality

It tells whether the final diagnosis is correct, but not whether the trajectory collected good evidence. As a result, the policy may fail to distinguish critical findings from noise and drift toward **indiscriminate information acquisition**.

SIFT Problem Setting

Instead of directly constructing ground-truth paths, SIFT turns how much **critical evidence** a trajectory retrieves into a reward signal.

SIFT: Main Figure



Components

Critical evidence scorer: marks diagnosis-relevant evidence at the sentence level.

Reward shaping: converts critical evidence retrieval into intermediate rewards for learning test selection.

SIFT: Offline Criticality Scoring

```
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EXAMPLE (illustrative only)
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ACTION_HISTORY:
[
  {
    "action": "Order comprehensive metabolic panel",
    "result": "Lipase returned at 1200 U/L. ALT is 85 U/L."
  },
  {
    "action": "Ask about alcohol use",
    "result": "Patient denies alcohol use."
  }
]

FACT_LIST:
[
  "Lipase 1,200 U/L",
  "ALT 85 U/L",
  "AST 72 U/L",
  "No alcohol use"
]

Output:
discovered_fact_indices: [0,1,3]
```

1. Fact extraction

Before training, atomic clinical facts are extracted from the patient record M.

$F = \phi(M) = \{f_1, \ldots, f_n\}$

Each f_i is a single observation, such as a lab value, imaging finding, or physical exam result, without mixed interpretation.

2. Criticality scoring

$w_f = \psi(f \mid d^*, o_0), \quad w_f \in \{0, 1, 2, 3\}$

Each fact is scored by how strongly it confirms the diagnosis, conditioned on the gold diagnosis d^* and initial presentation o_0.

Score meaning

- 0: irrelevant to the diagnosis
- 1: supportive but non-specific evidence
- 2: significant evidence
- 3: hallmark / pathognomonic finding

Scoring is performed offline, so ψ is not called during rollout.

SIFT: Method

1. Why is terminal reward insufficient?

A binary reward that only checks whether the final diagnosis is correct is observed only at the end of the episode. It does not indicate which action contributed to the correct diagnosis, so the agent may broadly collect diagnostically irrelevant information.

2. Critical recall

Measures the fraction of critical facts retrieved by the trajectory.

$$CR(\tau) = \frac{\sum_{f \in D(\tau)} w_f}{\sum_{f \in F} w_f}$$

F: full fact set, $D(\tau)$: discovered fact set, w_f: criticality score.

3. Criticality-weighted reward

The reward combines diagnostic correctness and critical evidence retrieval.

$$R(\tau) = \underbrace{(\alpha CR(\tau) + \beta)\,\mathrm{acc}(\hat{d}, d^*)}_{\text{correctness reward}} + \underbrace{\eta CR(\tau)}_{\text{partial credit}} - \underbrace{\lambda \frac{t}{T}}_{\text{step penalty}}$$

- **Correctness reward**: if correct, base reward β plus $\alpha CR(\tau)$; if incorrect, this term is 0
- **Partial credit**: even if incorrect, retrieving critical evidence gives $\eta CR(\tau)$
- **Step penalty**: penalizes using more test steps. t: used steps, T: maximum steps

Existing Trajectory-Level Approaches

LDTL

S initial	A	B	C	END accuracy

construct trajectory $z = (A, B, C)$

follow final confidence / accuracy

SIFT

S initial	A score 0	B score 3	C score 2	END $R(\tau) = 0 + 3 + 2$

measure how many critical facts are retrieved within trajectory $\tau = \{(a_t, o_t)\}$

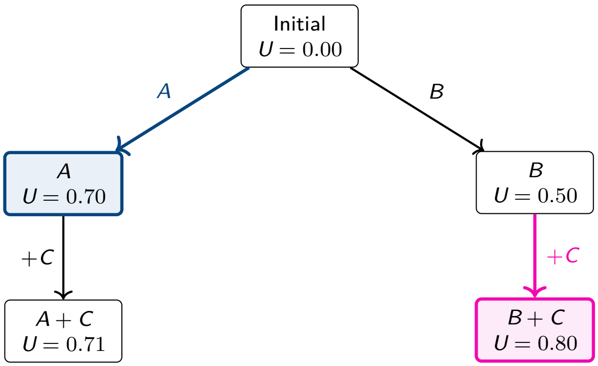
$R(\tau) = (\alpha CR + \beta)acc + \eta CR - \lambda t/T$

Shared Assumption

They implicitly treat the information provided by each test as relatively **independent**. In other words, knowing one test result is assumed not to substantially change the predictive value of adding another test.

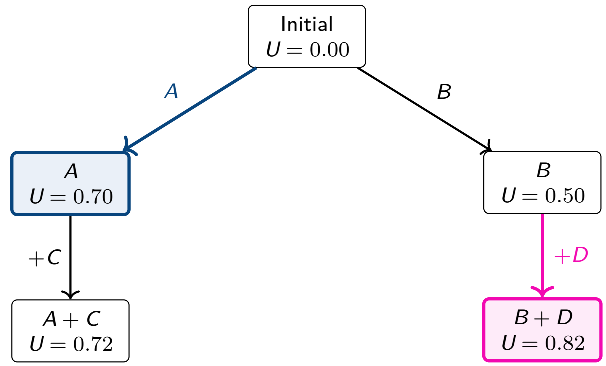
Greedy Does Not Fail Because Intermediate Reward Is Missing

Example 1: same second test C



Greedy selects A because $U(A) > U(B)$. But when the same C is added, $B + C$ becomes better.

Example 2: different second test



At first, A looks better, but B creates a higher trajectory utility when paired with D.

Takeaway

Even with intermediate rewards, evaluating A and B independently makes A look better. The issue is not where the reward is placed, but that the value of one test depends on **which later tests it is combined with**.

So What Can We Do?

1. Bayesian network

Description
Explicitly models conditional dependencies between tests and updates the posterior as evidence is added.

Pros

- Can handle combination effects and redundancy in principle
- Can define query order probabilistically

Cons

- Hard to convert natural-language reports into discrete variables e.g., CT report $\rightarrow$ severe: yes/no reduces expressiveness

2. *n*-step lookahead training

Description
Instead of evaluating only the current test, measure the utility of possible *n*-step combinations ahead and use it as a learning signal.

Pros

- Does not require labeling all subsets
- Can directly teach synergy in small combinations

Cons

- Computational cost grows exponentially

Max step	Subsets	Coverage
1	12	0.29%
2	78	1.90%
3	298	7.28%
4	793	19.37%
5	1,585	38.71%
6	2,509	61.27%
7	3,301	80.61%
8	3,796	92.70%

Thank You